

String Transformation: Minimum Operations

Problem Description:

Given two strings, source and target, write a REST service that calculates the minimum number of operations required to transform source into target. The allowed operations are:

- Insert a character.
- Delete a character.
- Replace a character.

Input:

JSON object containing two strings: source and target, each with a maximum length of 1,000.

Input will be sent on resource: **POST /calculate**

Output:

Return JSON object containing the list of actions and the minimum number of operations required.

Example Input:

```
{
  "source": "kitten",
  "target": "sitting"
}
```

Example Output:

```
{
  "actions_num": 3,
  "actions": [
    "Replace 'k' with 's'",
    "Replace 'e' with 'i'",
    "Insert 'g'"
  ]
}
```

Explanation:

Replace 'k' with 's' (kitten → sitten).

Replace 'e' with 'i' (sitten → sittin).

Insert 'g' (sittin → sitting).

Considerations:

- Not only the algorithm itself, but the project as a whole will be reviewed. Produce the quality you are used to at work.
- Include README if necessary, code comments where you think they should apply. Anything that helps the review process will be valued.
- You can improve the quality of the assignment by writing tests.
- And anything else that comes to your mind when delivering a project.